SCALER SCHOOL OF BUSINESS X IIT ROORKEE

AlgoVision: Data Analytics Case Competition

BY - KOED x COMET'26

Decoding Paytm's User Engagement Puzzle

EXECUTIVE SUMMARY

Paytm's churn stays close to 20% across user groups, showing that the issue is broader than any one segment. The main gap is weak repeat habit formation, with users often stopping after a single use case. The biggest opportunity lies in building deeper multi-feature usage and stronger return journeys.

MEET THE TEAM



Akshay Sinha
Product Management



Nirmay Bhavsar
Growth Strategy



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Marketing Strategy

50,000 Users Analyzed | 2.5-Year Window (Jan 2022 — Dec 2024) | 11 Variables | 8 App Features

Paytm has scale. The challenge is keeping users engaged.

The Super App Ecosystem

Payments & Utilities	UPI Payment	Wallet	Bill Payment	Recharge
Financial Products	Ticket Booking	Loan	Insurance	Investment

Four Core Business Challenges

- **Retention:** 1 in 5 users churns (20.1% churn rate)
- **Feature Adoption:** Users engage with only 1 feature each
- **Funnel Drop-offs:** Significant leakage at every stage
- **Transaction Failures:** 7.5% failure rate (mean success = 92.5%)

THE SCALE OF THE PROBLEM

~10,000

users lost to churn

Revenue at Risk

10,026 churned × 15 txns × Rs 2,525 avg value

Rs 37.9 Cr at risk

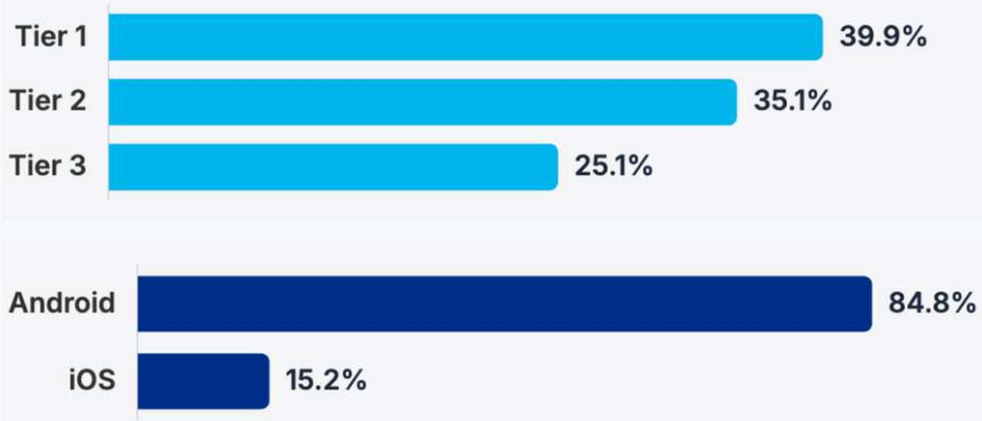
Key Question: What's driving churn, and which user segments should we prioritize for retention?

IIT ROORKEE

50,000 users. 11 variables. 2.5 years of data.

TOTAL USERS	MISSING VALUES	MONTHLY ACTIVE RATE	CHURN RATE	AVG TXN SUCCESS	AVG TXN VALUE
50,000	0	75.0%	20.1%	92.5%	₹2,525

Data Composition



Signup Range: 2022-01 to 2024-06
Last Active Range: 2024-01 to 2024-12

Data Quality Issues & Remediation

1. Date Paradox (Critical)

2,294 records (4.6%) have `last_active_date` before `signup_date`. Excluded from predictive modeling dataset.

2. MAU-Churn Paradox (Critical)

7,572 users (75.5% of all churned) are simultaneously `monthly_active=1` AND `churn_flag=1`. Treated churn as a forward-looking predictive label.

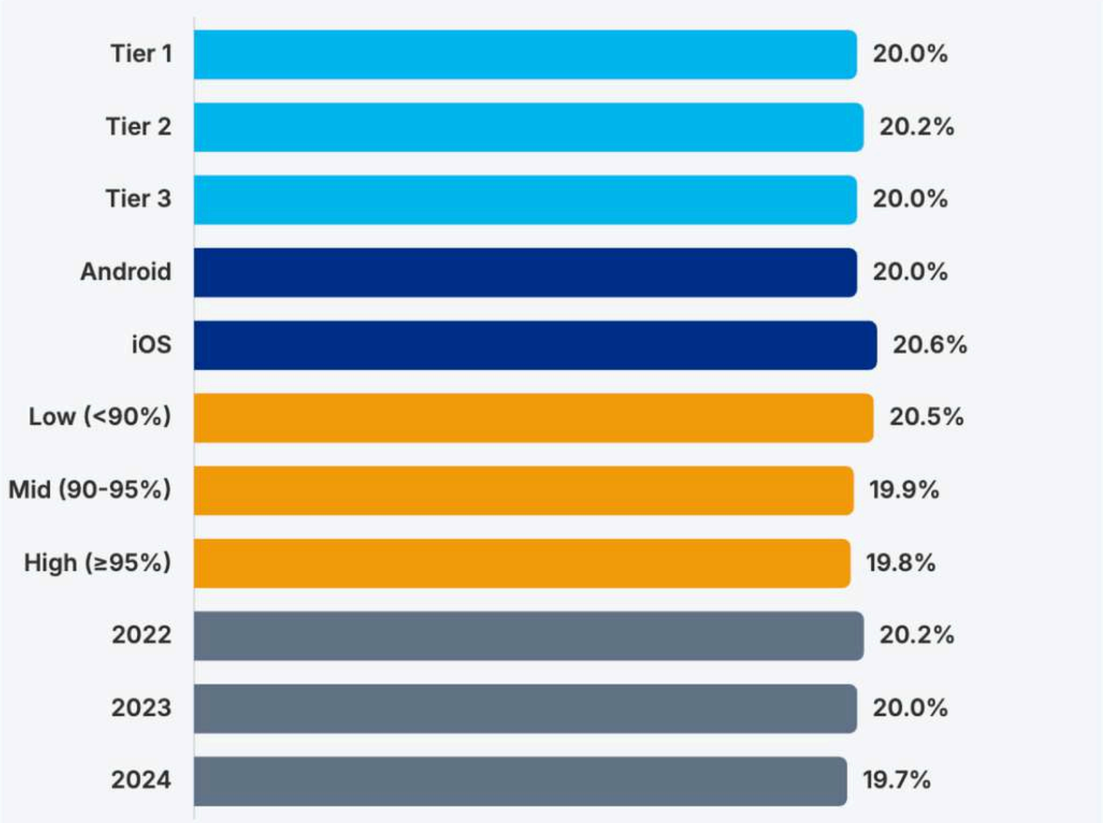
3. Uniform Feature Distribution (Note)

All 8 features hover evenly at **~12.5% each**. This confirms a synthetic dataset; methodology rigor prioritized over raw model accuracy.

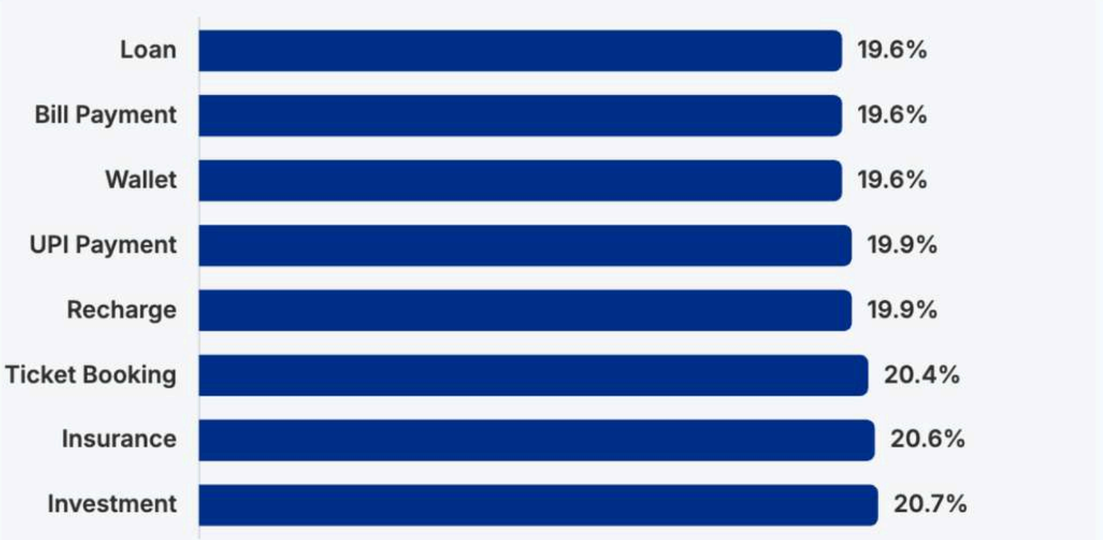
Takeaway: Clean dataset with two critical anomalies. After remediation, 47,706 records retained for analysis.

Churn is uniform at ~20% & no single factor stands out.

Churn across different user group



Churn by Feature Use

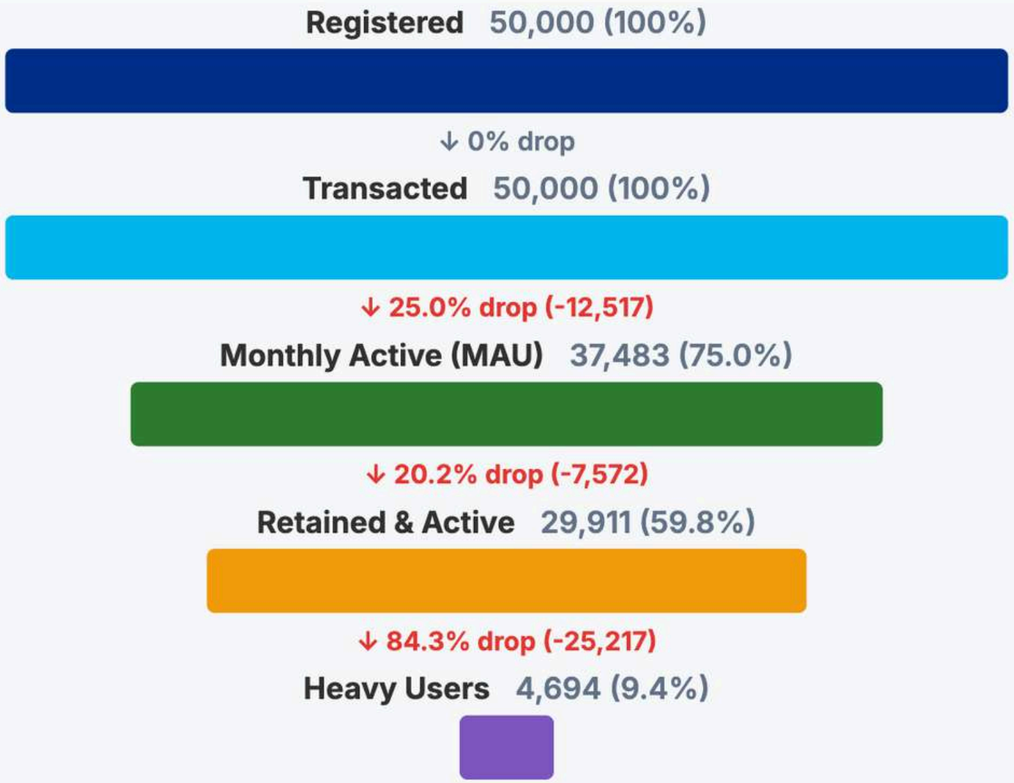


This is not limited to one audience type. The similar churn patterns suggest users are using Paytm for one task, not as a regular habit.

Takeaway : No single variable predicts churn. We need multi-variable risk scoring and segment-specific playbooks.

40% of registered users never become regular active users

Where users drop in the journey



Biggest user drop-offs

↘ Transacted → MAU	-12,517 users (25%)
↘ MAU → Retained Active	-7,572 users (20%)
↘ Retained → Power User	-25,217 users (84%)

The MAU-Churn Paradox

	Retained ✅	Churned ❌
Active (MAU=1)	**29,911** Healthy Core	**7,572** ⚠ 75.5% of churned users are still active. Most salvageable.
Inactive (MAU=0)	**10,063** zzz The "dormant middle". Highest-ROI re-engagement targets.	**2,454** Lost Causes

Takeaway: The biggest opportunity is the 10,063 dormant-but-retained users who haven't churned yet.

Six RFM segments reveal who to prioritize for retention.

CHAMPIONS
7,567
15.9% of users
19.8% Churn
High-frequency, high-value, recently active.

LOYAL
5,933
12.4% of users
20.3% Churn
High frequency, lower value, recent.

RECENT USERS
11,180
23.4% of users
20.2% Churn
Low frequency, recently joined/active.

AT RISK
11,141
23.3% of users
19.9% Churn
High frequency but going dormant.

CAN'T LOSE
5,910
12.4% of users
19.8% Churn
High value but disengaging rapidly.

HIBERNATING
5,975
12.5% of users
20.7% Churn
Low everything, completely dormant.

Priority Action Matrix

SEGMENT (% OF BASE)	STRATEGY	PRIORITY
At Risk (23.3%)	Win-back campaigns, reactivation nudges	P0
Can't Lose (12.4%)	VIP treatment, exclusive features, personal outreach	P0
Recent Users (23.4%)	Onboarding optimization, feature discovery, habit loops	P1
Champions (15.9%)	Referral programs, upsell to financial products	P2

K-Means Validation (K=7)

Unsupervised K-Means clustering independently confirms RFM findings. **All 7 clusters show churn rates within ±0.7pp of 20%**, reinforcing that churn is platform-wide, not segment-specific.



Takeaway: "At Risk" + "Can't Lose Them" = 35.7% of users. These are the highest-priority retention targets.

7.5% failure rate = ~1.1 failed transactions per user. Every failure erodes trust.

AVG TXNS/USER	AVG TXN VALUE
15.0	₹2,525

SUCCESS RATE	TOTAL FAILED TXNS
92.5%	~56,000

Revenue at Risk Calculation

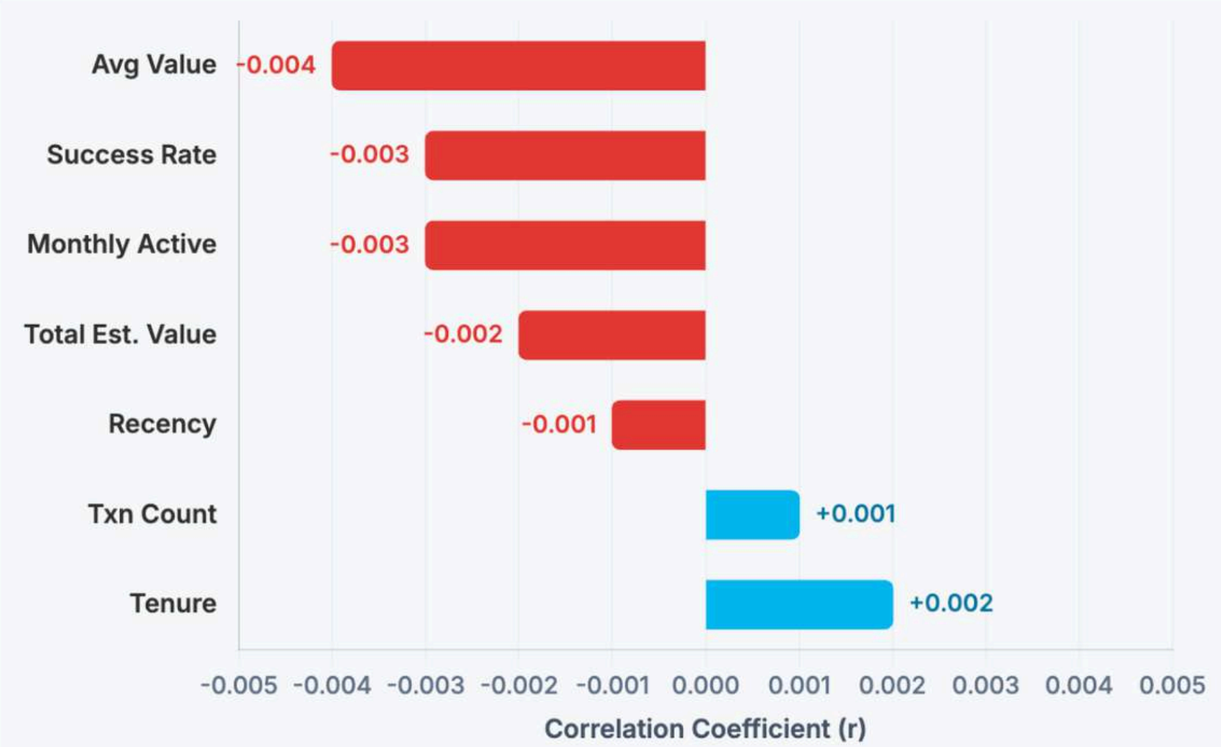
10,026 churned × 15 txns × ₹2,525

₹37.97 Cr at risk

Recovering just 10% = ₹3.8 Cr saved

Correlation with Churn (Point-Biserial)

All |r| < 0.01: No linear relationship with churn exists in this dataset.



Takeaway: Improving the success rate from 92.5% → 95% eliminates ~18,750 failed transactions = ₹4.7 Cr in recovered value.

Five strategic bets to reduce churn and grow engagement.

P0 — HIGH IMPACT

Fix Transaction Success Rate

PROBLEM

7.5% failure rate = ~56K failed txns. Every failure is a trust-erosion event.

SOLUTION

Invest in infra reliability, smart retry logic, real-time monitoring, and proactive user notifications.

SUCCESS METRICS

18,750 failures, ₹4.7 Cr saved. KPI: Success Rate ≥ 95%

P0 — HIGH IMPACT

Reactivate the "Dormant Middle"

PROBLEM

10,063 retained but inactive users are disengaged and at immediate risk of churn.

SOLUTION

Targeted re-engagement campaigns via personalized push notifications and seasonal offers.

SUCCESS METRICS

10K reactivation targets KPI: +20% MAU uplift

P1 — MEDIUM TERM

Feature Cross-Pollination

PROBLEM

Users engage with only 1 feature. Low switching costs and no ecosystem lock-in.

SOLUTION

Contextual discovery surfaces and cross-feature incentives (e.g., use 3+ features, earn cashback).

SUCCESS METRICS

2× LTV for multi-feature adopters. KPI: Avg features/user ≥ 2.5

P1 — MEDIUM TERM

Tier 3 Expansion

PROBLEM

Only 25.1% of users are Tier 3, despite massive untapped market population.

SOLUTION

Localized vernacular onboarding, lightweight app version, and regional retailer partnerships.

SUCCESS METRICS

2× addressable market base KPI: Tier 3 share ≥ 35%

STRATEGIC BUILD

Data-First Churn Prevention System

PROBLEM

Current data cannot predict churn (AUC ~0.50). Team is flying blind on retention.

SOLUTION

Instrument session-level events and NPS. Build a real-time composite risk score with automated nudges.

SUCCESS METRICS

Target AUC > 0.75 | ₹10 Cr+ value KPI: Churn rate < 15%

Churn is bigger than any one feature, it's about the full user journey.

No single dimension explains why 1 in 5 users leave. The solution isn't better targeting, it's **better infrastructure, richer data, and segment-specific playbooks**.

REVENUE AT RISK	DORMANT TO REACTIVATE	STRATEGIC BETS	QUICK WIN (FIX TXNS)
₹37.9 Cr	10,063	5	₹4.7 Cr

Action Plan Priority Summary

- P0 Fix Success Rate
- P0 Reactivate Dormant Users
- P1 Cross-Feature Nudges
- P1 Tier 3 Expansion
- STRATEGIC Data-First Churn System

Funnel Drop-offs, Heavy Users, and “Lost Causes”

Q1) How is the drop-off from MAU → Retained & Active calculated?

This is a simple subtraction, not a prediction.

Monthly Active (MAU) = users where monthly_active = 1 = 37,483

Retained & Active = users where monthly_active = 1 and churn_flag = 0 = 29,911

Drop-off

37,483 – 29,911 = 7,572 users

These 7,572 are users who are:

active this month, but still marked churned / at-risk

So the “drop” is simply:

users active now but not considered safely retained

Q2) What defines Heavy Users?

Heavy Users (earlier Power Users) are users who satisfy both:

Transactions ≥ 20

Success rate ≥ 95%

This is above the dataset average:

Avg txns = 15

Heavy users = 20+

Total Heavy Users

4,694 users (9.4%)

Simple meaning:

highly engaged + reliable transactors

Q3) What are “Lost Causes”?

This is not an official segment, so don’t use this wording in the final deck.

Better phrase:

Fully churned inactive users

These are users with:

monthly_active = 0

churn_flag = 1

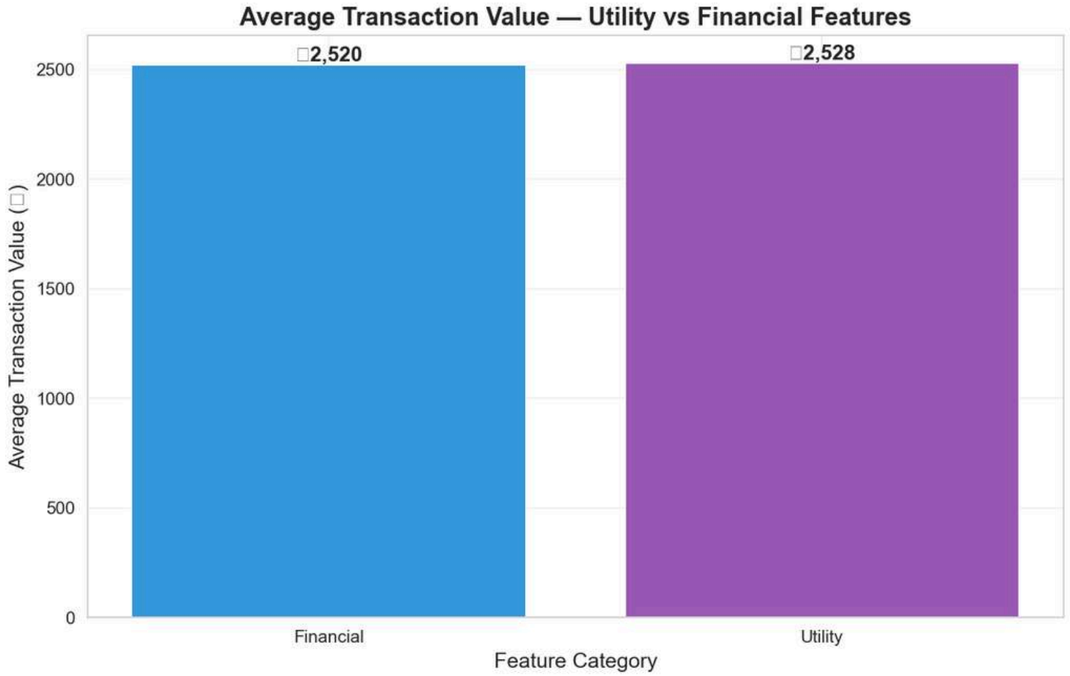
Count

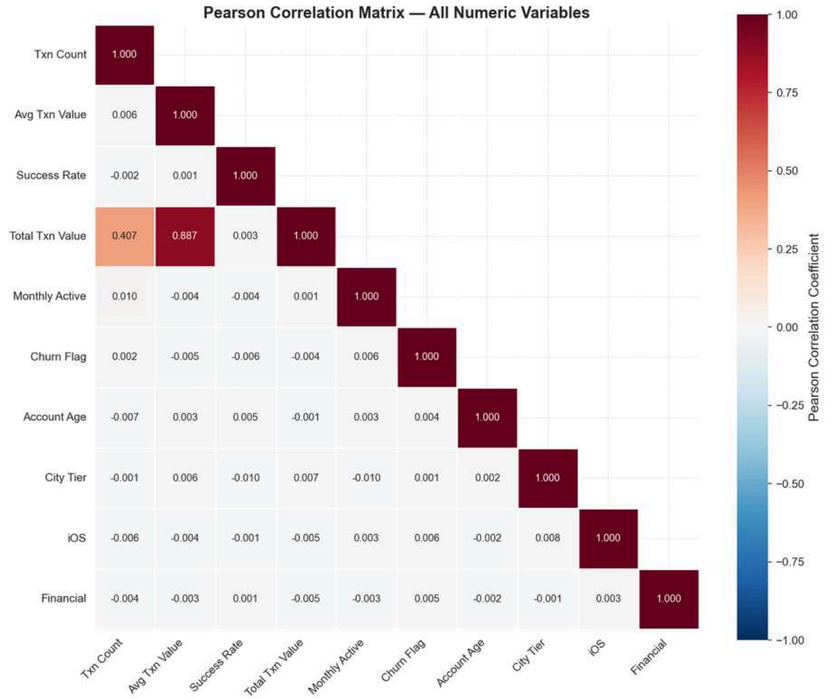
2,454 users

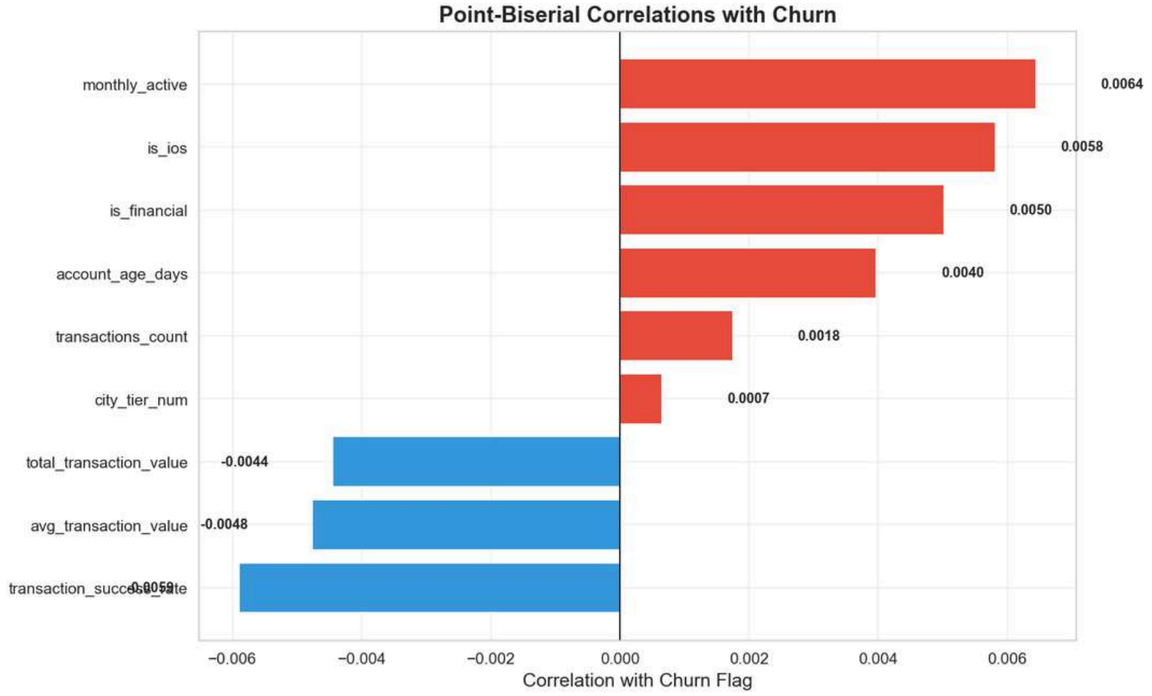
Meaning:

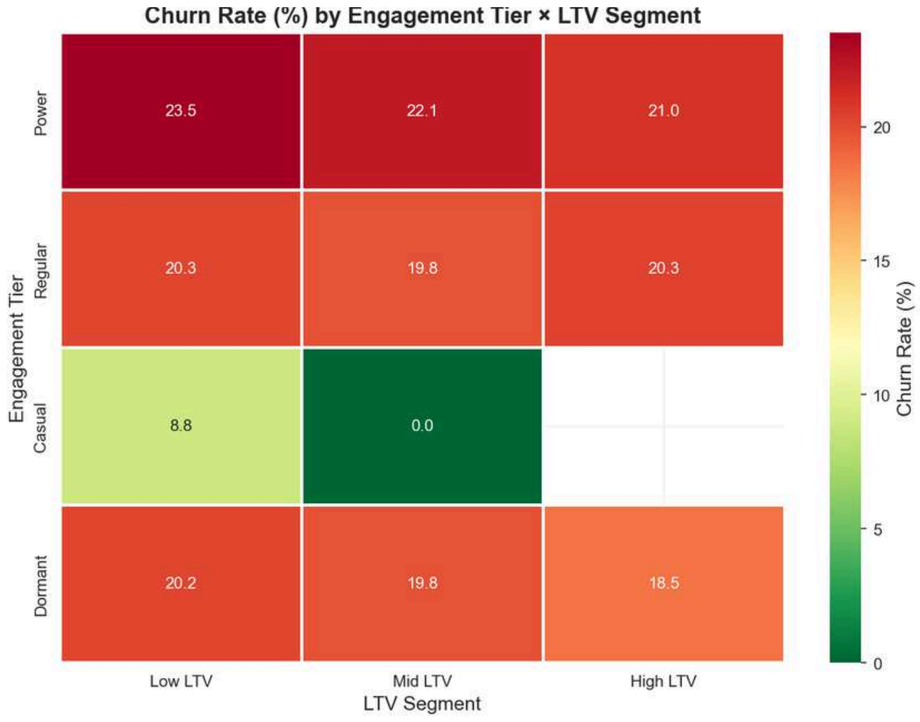
inactive and already lost, hardest to recover

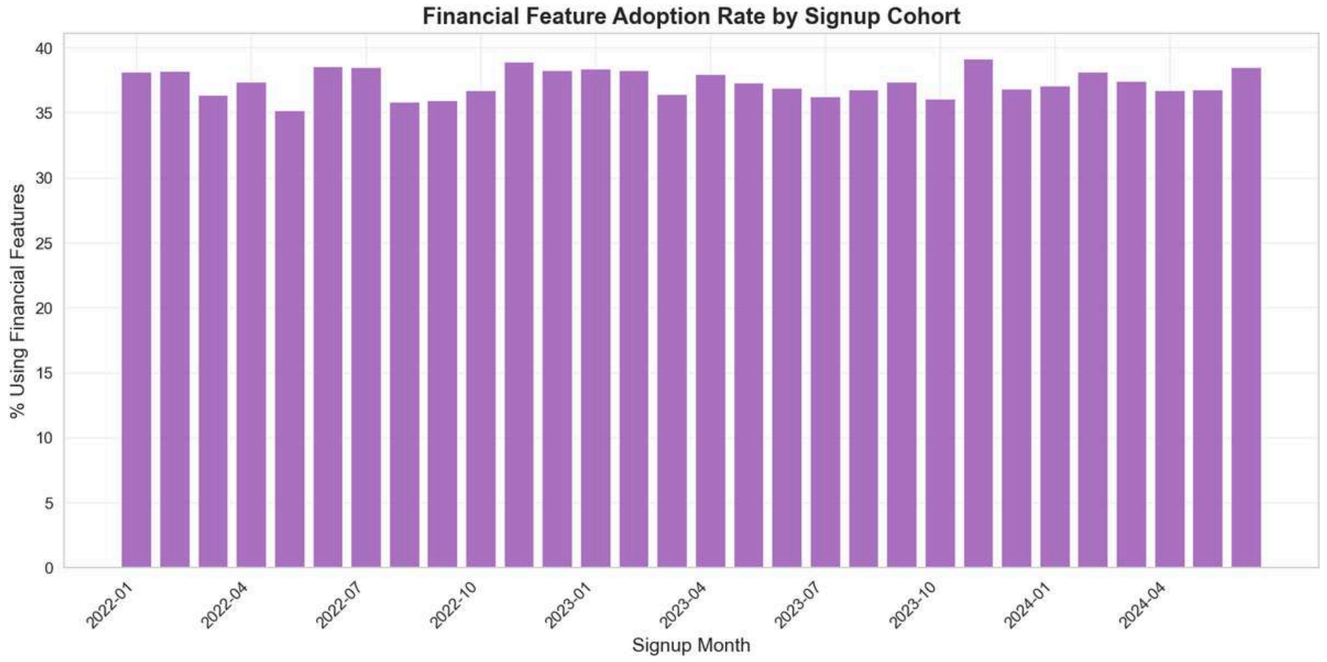
RFM Segmentation Q4) What exact metrics were used for segmentation? We used RFM = Recency + Frequency + Monetary Recency (R) Days since last activity Range = 1 to 365 days Frequency (F) Transaction count Range = 2 to 37 Monetary (M) Average transaction value Range = ₹50 to ₹5,000 Each was divided into quartiles (1–4 score). So: top 25% = score 4 bottom 25% = score 1 This made the segmentation data-based, not random. Q5) What does “high frequency” actually mean? This is the most common judge question. Exact threshold Top quartile ≈ 20+ transactions So: F = 4 → 20+ txns F = 3 → 15–20 F = 2 → 10–15 F = 1 → 2–10 So yes: high frequency = roughly 20+ transactions Q6) Exact rules for each segment Champions RFM total ≥ 10 high recency + high frequency + high spend Loyal R ≥ 3 and F ≥ 3 recent + frequent, moderate spend Recent Users R ≥ 3 and F ≤ 2 recently active but low frequency At Risk R ≤ 2 and F ≥ 3 used often before, now going dormant Can’t Lose R ≤ 2 and M ≥ 3 high spenders becoming inactive Hibernating Everything else low recency + low frequency + low spend

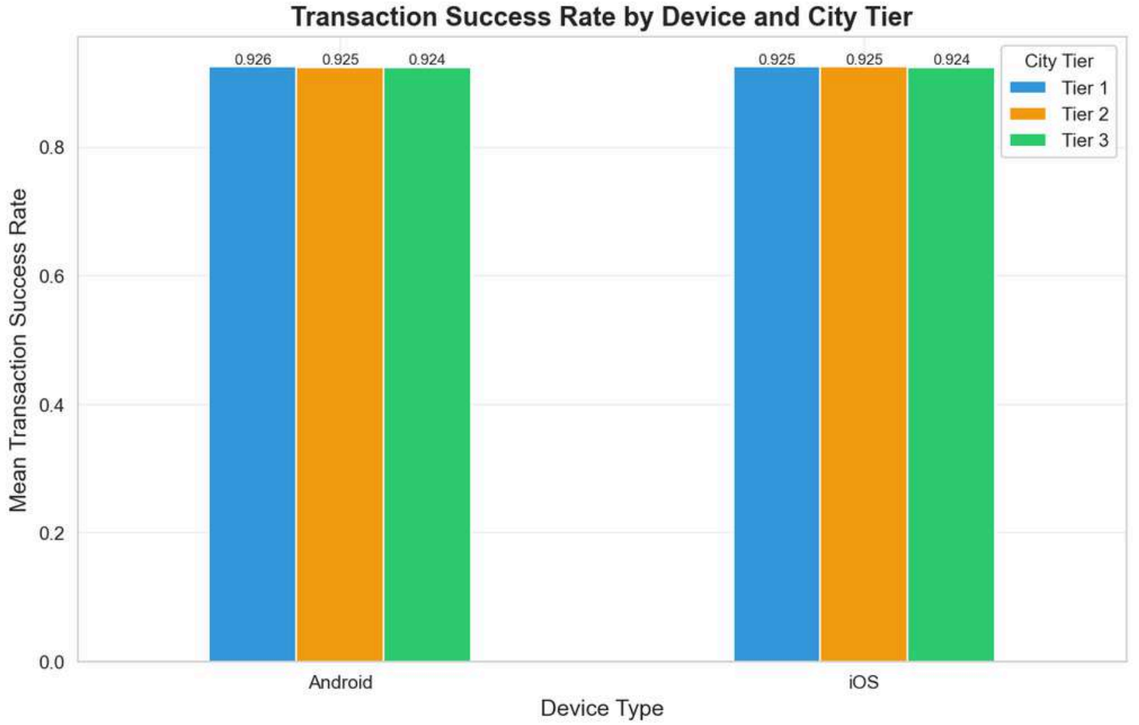

Average Transaction Value — Utility vs Financial Features
□2,520
□2,528
Average Transaction Value (□)
2500
2000
1500
1000
500
0
Financial
Utility
Feature Category

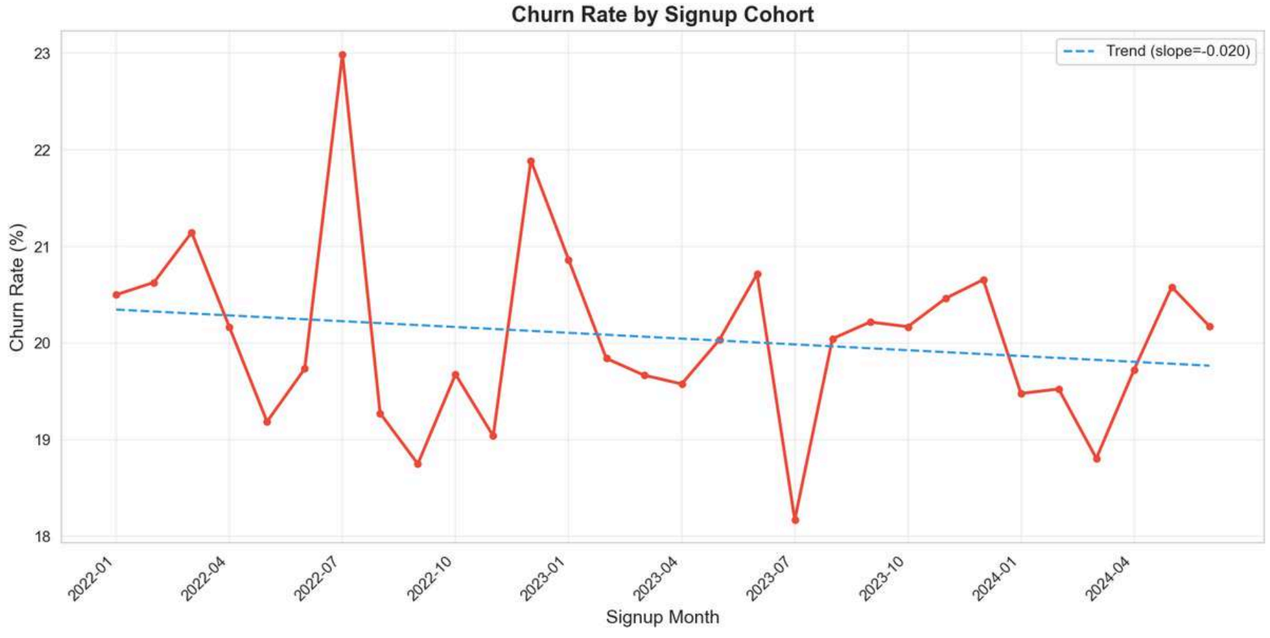

Pearson Correlation Matrix — All Numeric Variables
Txn Count
Avg Txn Value
Success Rate
Total Txn Value
Monthly Active
Churn Flag
Account Age
City Tier
iOS
Financial
Pearson Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

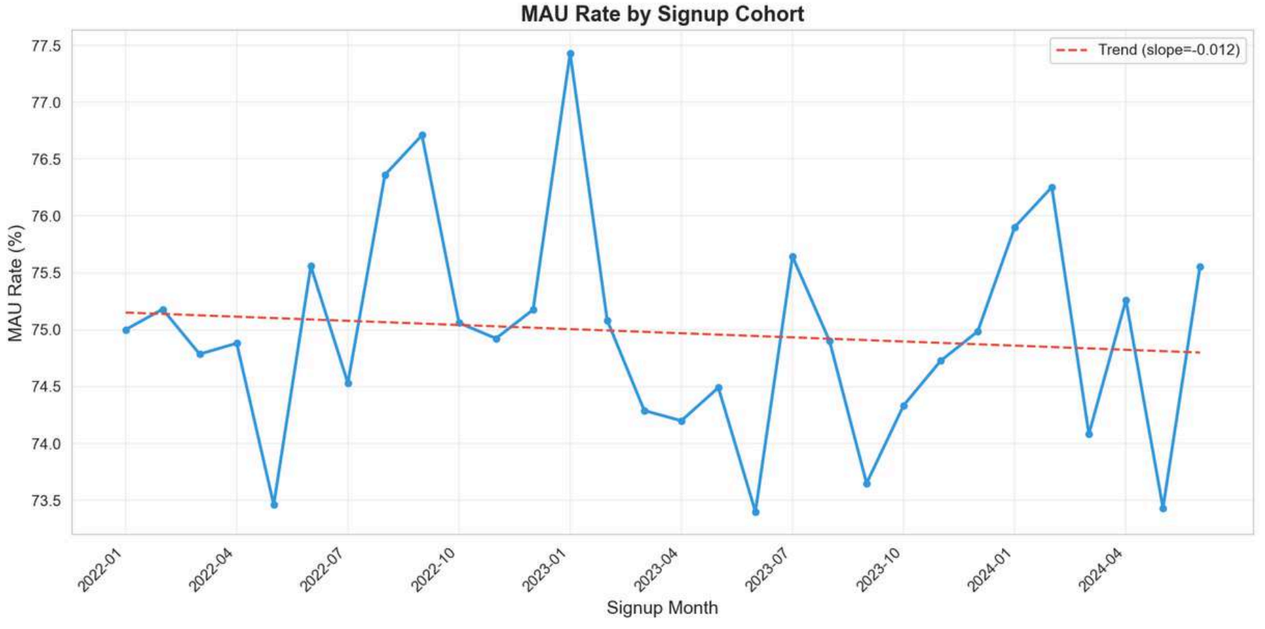

Point-Biserial Correlations with Churn
monthly_active
is_ios
is_financial
account_age_days
transactions_count
city_tier_num
total_transaction_value
avg_transaction_value
transaction_success_rate
0.0064
0.0058
0.0050
0.0040
0.0018
0.0007
-0.0044
-0.0048
−0.006
−0.004
−0.002
0.000
0.002
0.004
0.006
Correlation with Churn Flag

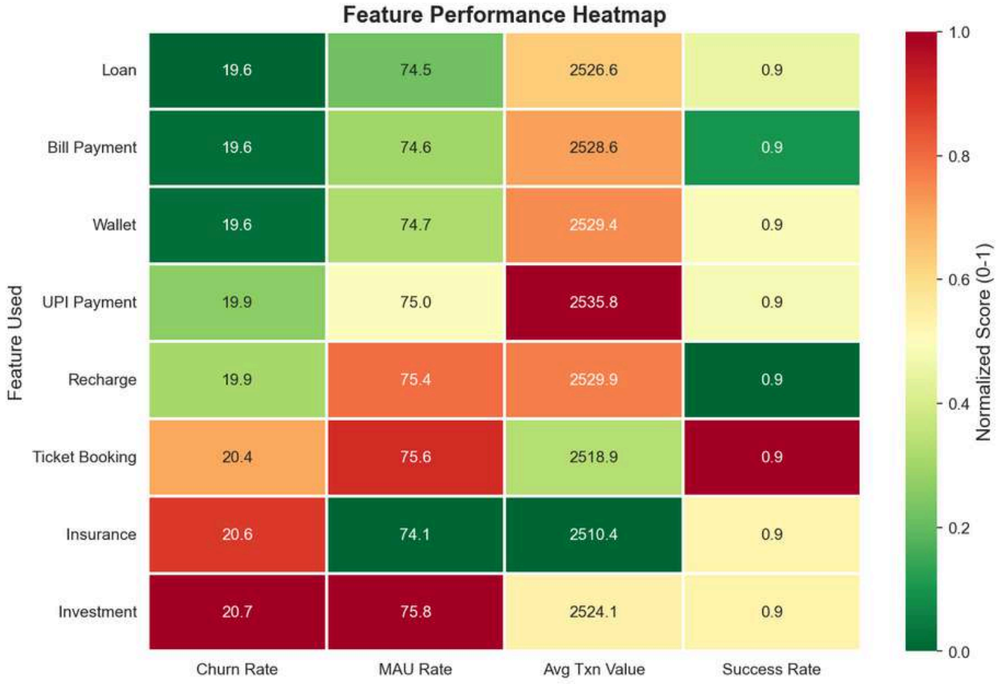

Churn Rate (%) by Engagement Tier × LTV Segment
Power
Regular
Casual
Dormant
Engagement Tier
23.5
22.1
21.0
20.3
19.8
20.3
8.8
0.0
20.2
19.8
18.5
Low LTV
Mid LTV
High LTV
LTV Segment
Churn Rate (%)
20
15
10
5
0

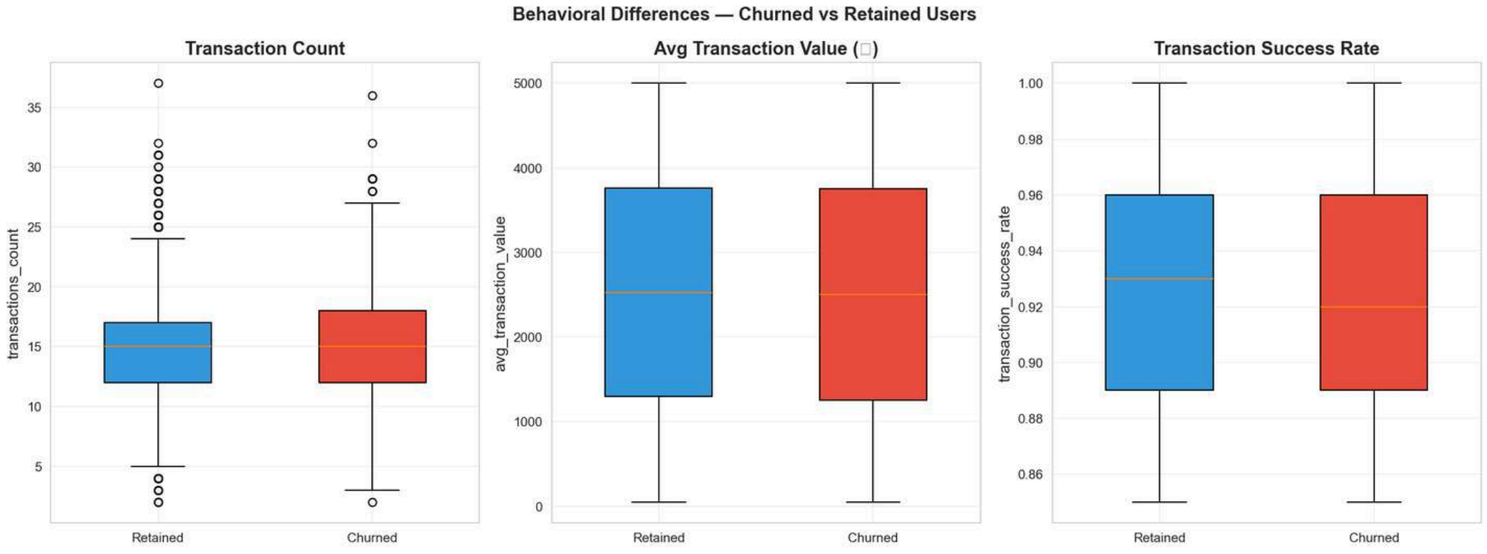

Financial Feature Adoption Rate by Signup Cohort
% Using Financial Features
Signup Month
0
5
10
15
20
25
30
35
40
2022-01
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04

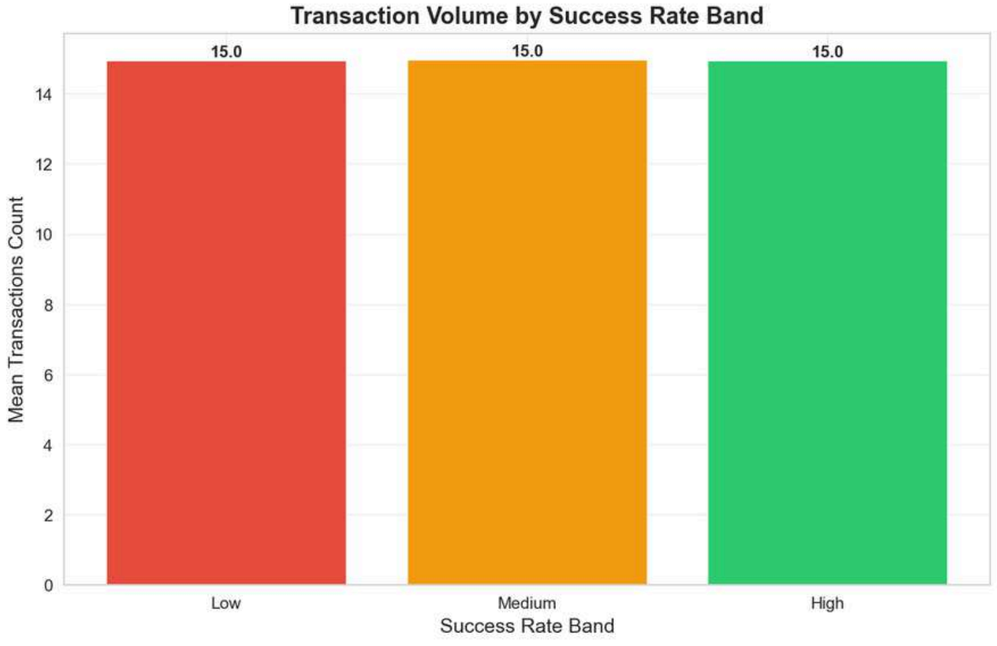

Transaction Success Rate by Device and City Tier
Mean Transaction Success Rate
Device Type
Android
iOS
0.926
0.925
0.924
0.925
0.925
0.924
City Tier
Tier 1
Tier 2
Tier 3
0.0
0.2
0.4
0.6
0.8

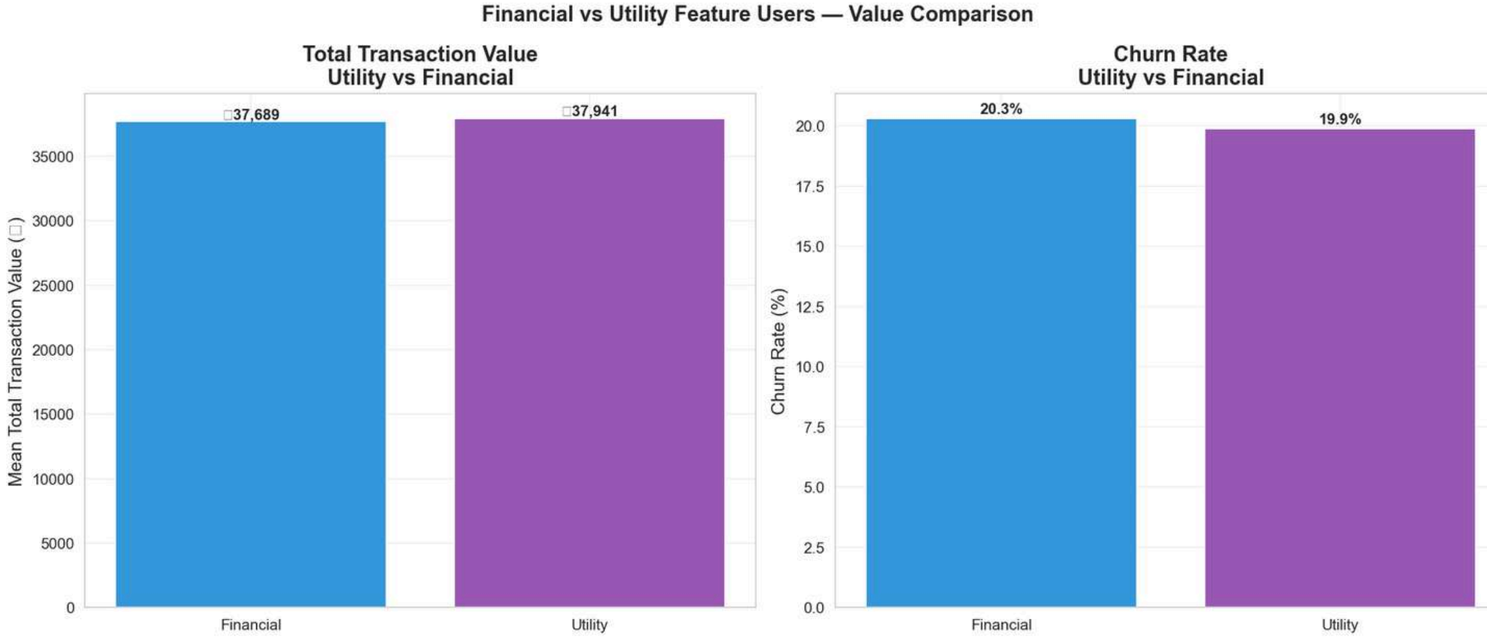

Churn Rate by Signup Cohort
Churn Rate (%)
Signup Month
Trend (slope=-0.020)
18
19
20
21
22
23
2022-01
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04

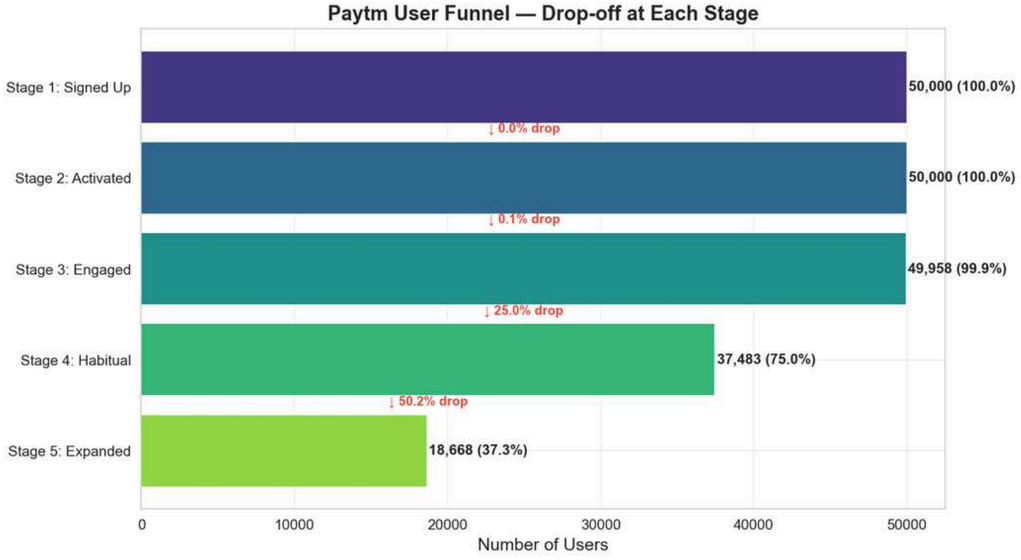

MAU Rate by Signup Cohort
MAU Rate (%)
Signup Month
Trend (slope=-0.012)
73.5
74.0
74.5
75.0
75.5
76.0
76.5
77.0
77.5
2022-01
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04

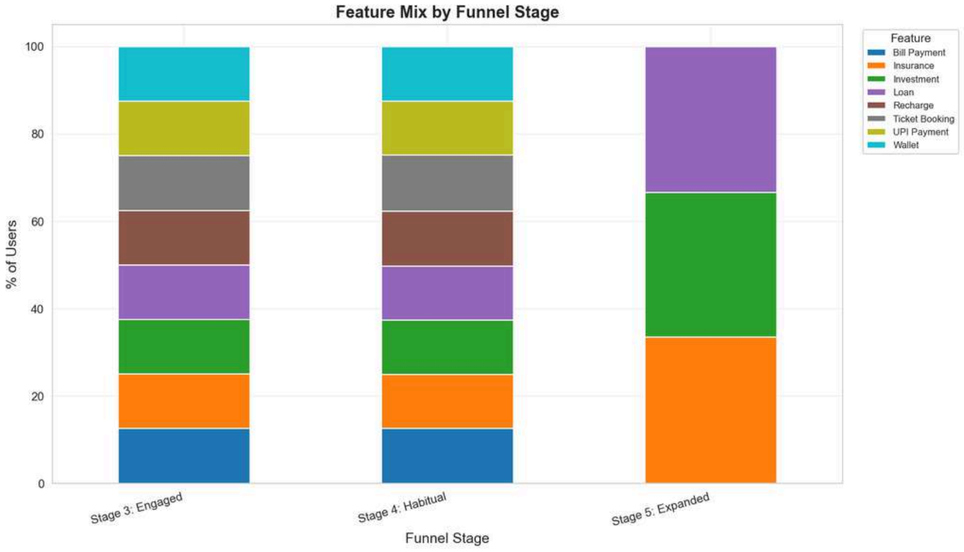

Feature Performance Heatmap
Feature Used
Loan
Bill Payment
Wallet
UPI Payment
Recharge
Ticket Booking
Insurance
Investment
19.6 74.5 2526.6 0.9
19.6 74.6 2528.6 0.9
19.6 74.7 2529.4 0.9
19.9 75.0 2535.8 0.9
19.9 75.4 2529.9 0.9
20.4 75.6 2518.9 0.9
20.6 74.1 2510.4 0.9
20.7 75.8 2524.1 0.9
Churn Rate
MAU Rate
Avg Txn Value
Success Rate
Normalized Score (0-1)
1.0
0.8
0.6
0.4
0.2
0.0

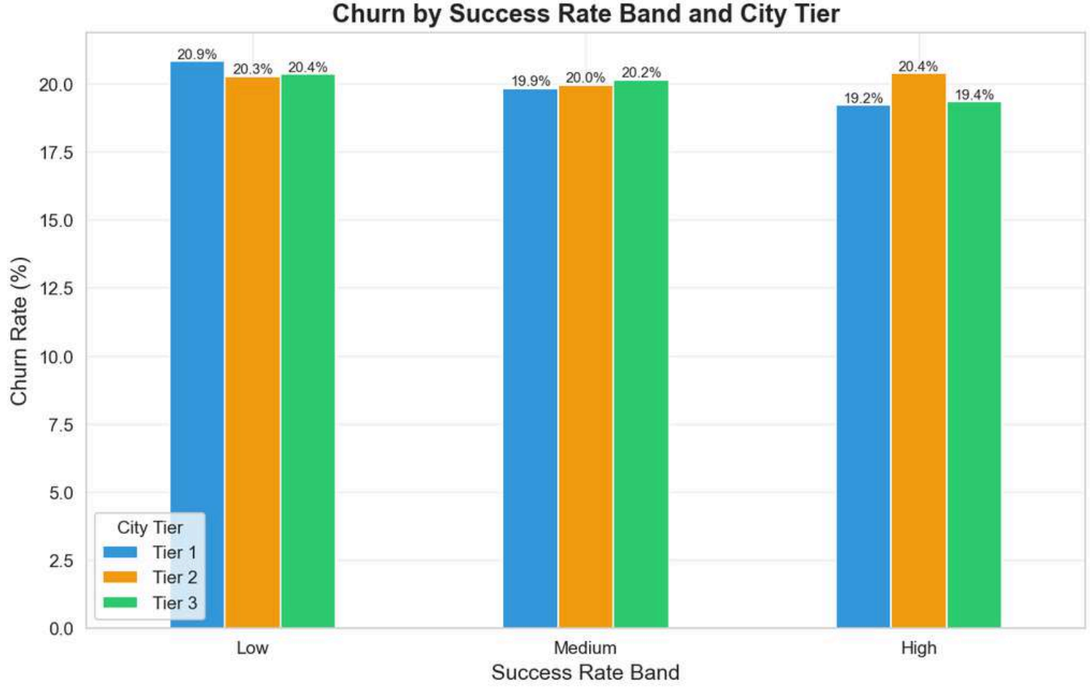

Behavioral Differences — Churned vs Retained Users
Transaction Count
transactions_count
Retained
Churned
Avg Transaction Value (□)
avg_transaction_value
Retained
Churned
Transaction Success Rate
transaction_success_rate
Retained
Churned

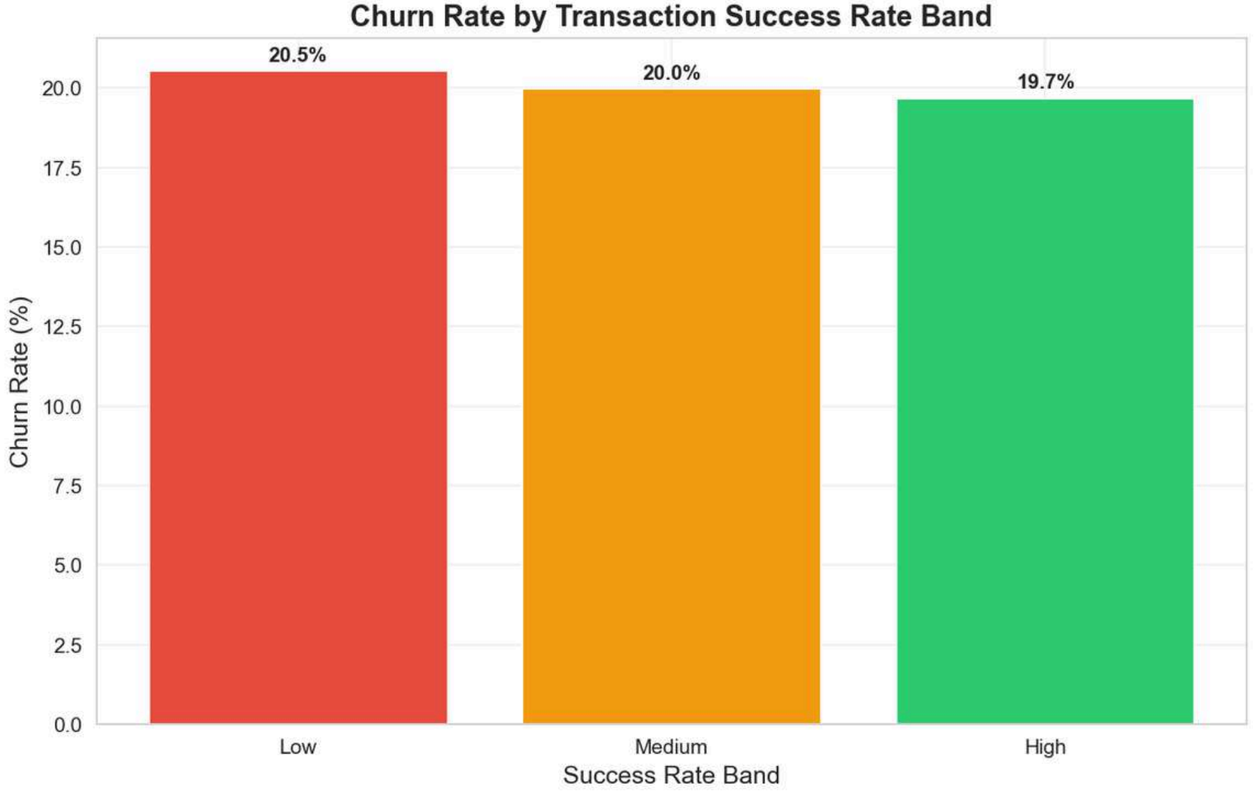

Transaction Volume by Success Rate Band
Mean Transactions Count
15.0
15.0
15.0
Low
Medium
High
Success Rate Band

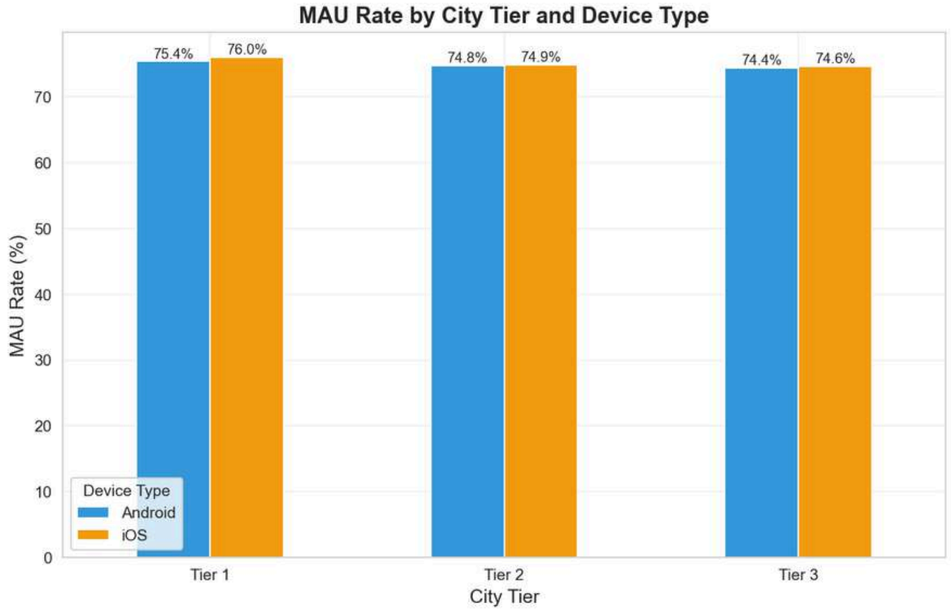

Financial vs Utility Feature Users — Value Comparison
Total Transaction Value
Utility vs Financial
Mean Total Transaction Value (□)
□37,689
□37,941
Financial
Utility
Churn Rate
Utility vs Financial
Churn Rate (%)
20.3%
19.9%
Financial
Utility

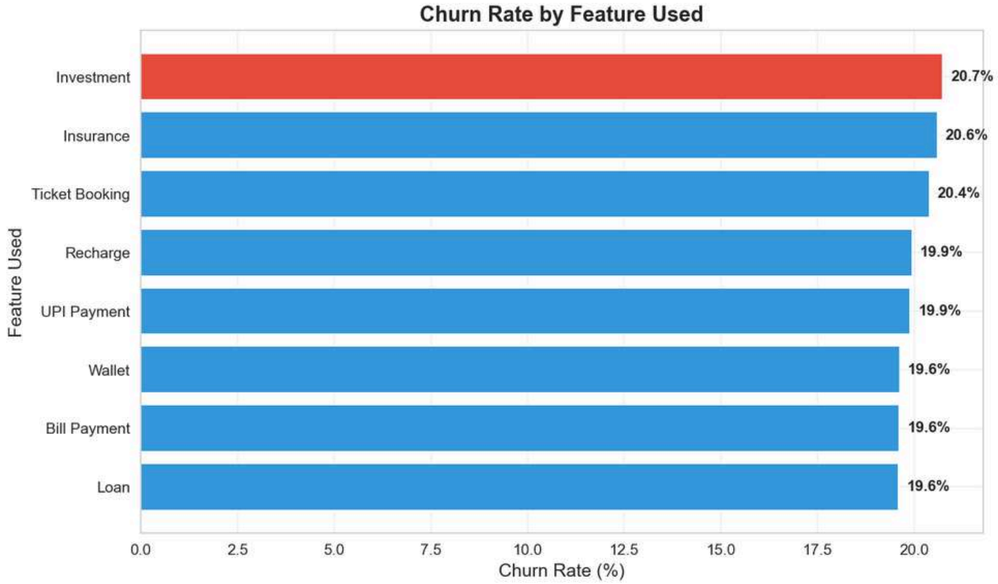

Paytm User Funnel — Drop-off at Each Stage
Stage 1: Signed Up
50,000 (100.0%)
↓ 0.0% drop
Stage 2: Activated
50,000 (100.0%)
↓ 0.1% drop
Stage 3: Engaged
49,958 (99.9%)
↓ 25.0% drop
Stage 4: Habitual
37,483 (75.0%)
↓ 50.2% drop
Stage 5: Expanded
18,668 (37.3%)
0
10000
20000
30000
40000
50000
Number of Users

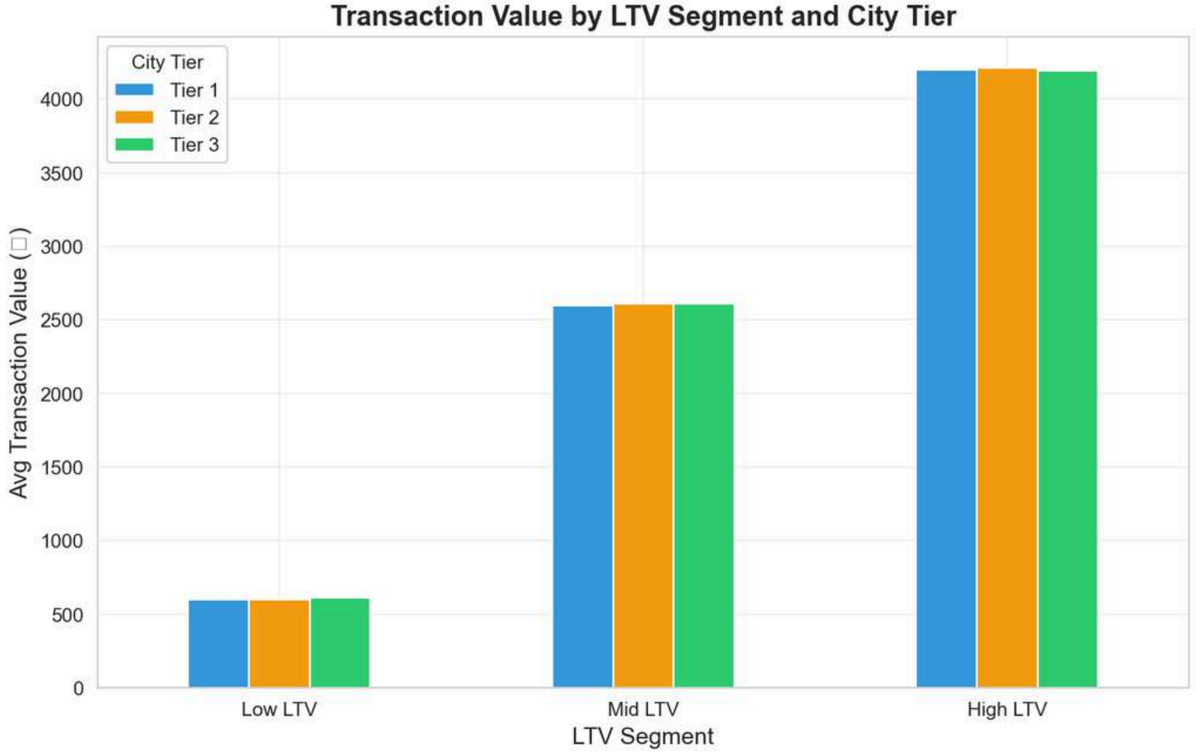

Feature Mix by Funnel Stage
% of Users
Feature
Bill Payment
Insurance
Investment
Loan
Recharge
Ticket Booking
UPI Payment
Wallet
Stage 3: Engaged
Stage 4: Habitual
Stage 5: Expanded
Funnel Stage

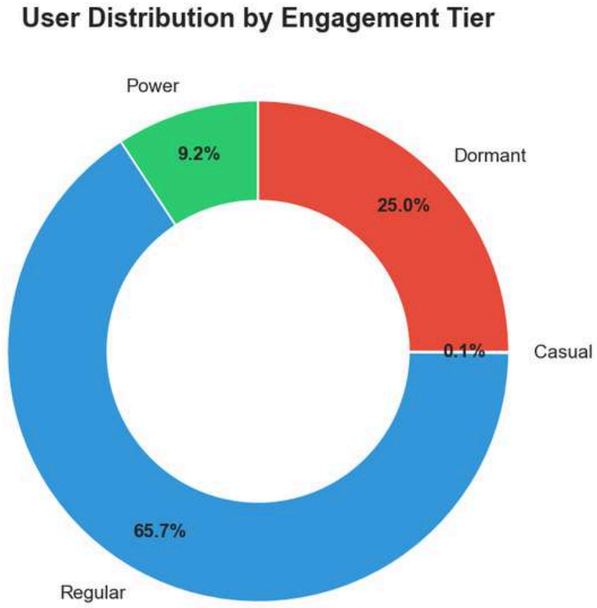

Churn by Success Rate Band and City Tier
Churn Rate (%)
20.9%
20.3%
20.4%
19.9%
20.0%
20.2%
19.2%
20.4%
19.4%
City Tier
Tier 1
Tier 2
Tier 3
Low
Medium
High
Success Rate Band

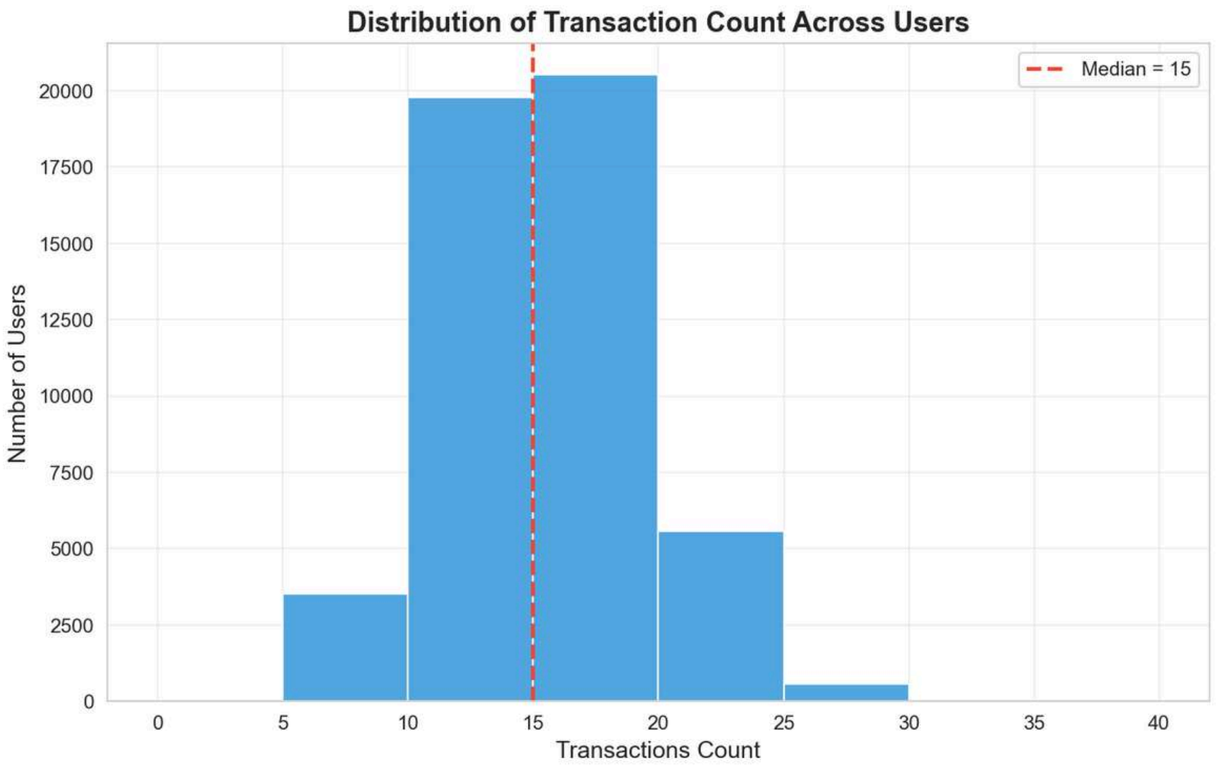

Churn Rate by Transaction Success Rate Band
20.5%
20.0%
19.7%
Churn Rate (%)
Low
Medium
High
Success Rate Band

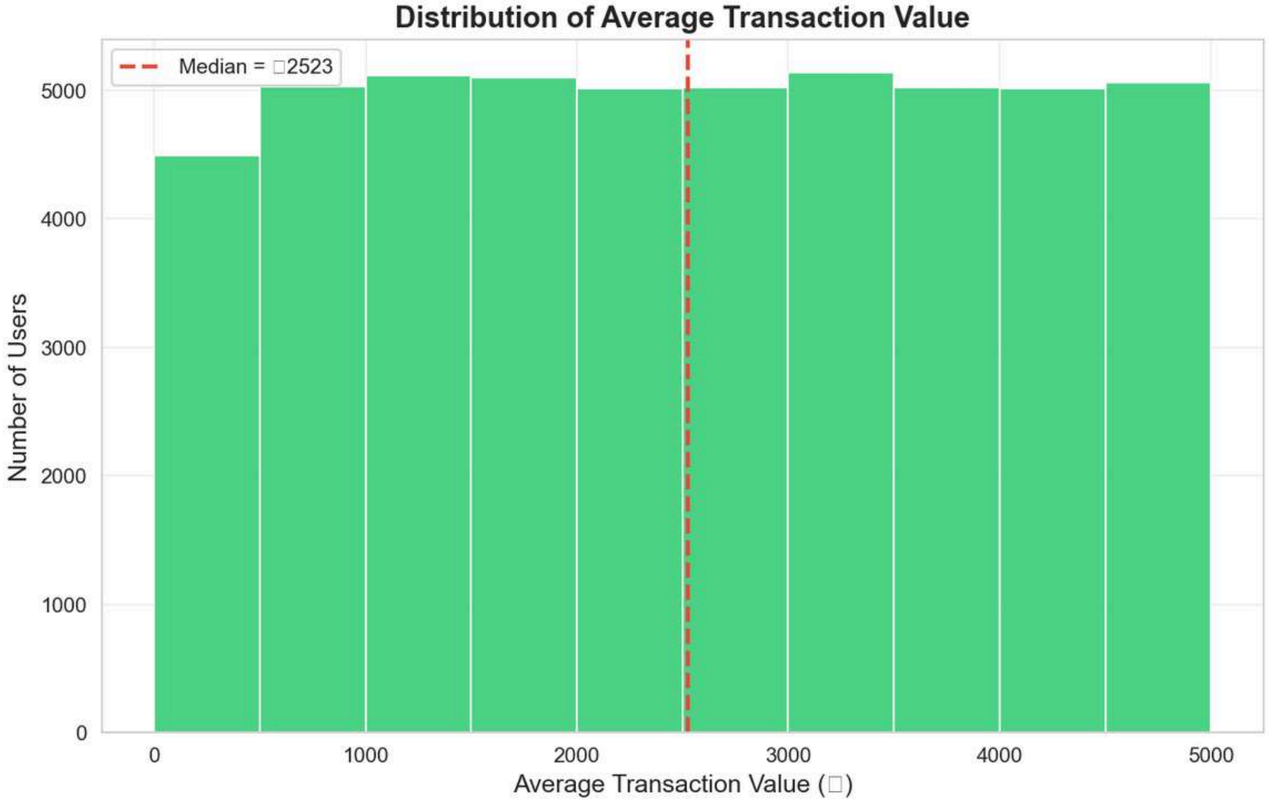

MAU Rate by City Tier and Device Type
MAU Rate (%)
City Tier
Tier 1
Tier 2
Tier 3
75.4%
76.0%
74.8%
74.9%
74.4%
74.6%
Device Type
Android
iOS


Churn Rate by Feature Used
Feature Used
Churn Rate (%)
Investment
Insurance
Ticket Booking
Recharge
UPI Payment
Wallet
Bill Payment
Loan
20.7%
20.6%
20.4%
19.9%
19.9%
19.6%
19.6%
19.6%


Transaction Value by LTV Segment and City Tier
Avg Transaction Value (□)
LTV Segment
Low LTV
Mid LTV
High LTV
City Tier
Tier 1
Tier 2
Tier 3


User Distribution by Engagement Tier
Power
9.2%
Dormant
25.0%
Casual
0.1%
Regular
65.7%


Distribution of Transaction Count Across Users
Median = 15
Number of Users
Transactions Count
0
2500
5000
7500
10000
12500
15000
17500
20000
0
5
10
15
20
25
30
35
40


Distribution of Average Transaction Value
Median = □2523
Number of Users
Average Transaction Value (□)
0
1000
2000
3000
4000
5000
0
1000
2000
3000
4000
5000

Models used:

Logistic Regression: AUC 0.506
Random Forest: AUC 0.507 (best)
Gradient Boosting: AUC 0.504